Pack Insert
Date : 28JUNE22
Date : 23NOV22*
Date : 24FEB24**
Malaysia Export
(Ambernath)

For the use only of a Registered Medical Practitioner or a Hospital.

Polymyxin B For Injection U.S.P.

POLY-MxB

500,000 Units

(Lyophilized)

For Intrathecal / I.M. / I.V. Infusion

DESCRIPTION:
Polymyxin B Sulphate is one of a group of basic polypeptide antibiotics derived from Bacillus polymyxa. **POLY-MxB** is a white or almost white cake/powder which in reconstitution with 5ml water for injection gives clear colourless liquid. The Lyophilized cake form is suitable the Lyophilized cake form suitable for preparation of sterile solutions for intramuscular, intravenous drip, intrathecal only (see Dose & Administration section below).

COMPOSITION:
Each vial contains:
Polymyxin B Sulphate U.S.P. Equivalent to Polymyxin B ……… 500,000 Units.

CLINICAL PHARMACOLOGY:
PHARMACODYNAMICS:
Polymyxin B Sulfate is a mixture of polymyxins B1 and B2, obtained from Bacillus polymyxa strains. They are basic polypeptides of about eight amino acids and have cationic detergent action on cell membranes. Polymyxin B is used for infections with gram – negative organisms, but may be neurotoxic and nephrotoxic. All gram- positive bacteria, fungi, and the gram – negative cocci, N. *gonorrhea* and N. *menigitidies*, are resistant.

PHARMACOKINETICS:
Polymyxin B Sulfate is not absorbed from the normal alimentary tract. Since the drug loses 50 percent of its activity in the presence of serum, active blood levels are low. Repeated injections may give a cumulative effect. Levels tend to be higher in infants and children. The drug is excreted slowly by the kidneys. Tissue diffusion is poor and the drug does not pass the blood brain barrier into the cerebrospinal fluid. In therapeutic dosage, Polymyxin B sulfate causes some nephrotoxicity with tubule damage to a slight degree.
Polymyxin B is administered as its sulfate salt which is the active antibiotic. Thus conversion is not required. It is eliminated mainly non-rental pathways. As a result, Polymyxin B concentration in the urine is low. The total body clearance appears to be relatively insensitive to renal function. Inter- individual variability in plasma concentration is low. Its serum half-life depends on the age of the patient and ranges from 3.1 (in neonates) to 13.6 hours. The unbound fraction in plasma is approximately 40%. A loading does not seems to improve significantly its PK parameters.

Pharmacokinetic – Pharmacodynamics relationship:
Tissue binding is a dominant factor in the distribution and elimination of polymyxins. Old experimental studies have shown that polymyxins accumulate in liver, lung, kidney, heart, and muscles. At 24h, less than 50% of the doses of Polymyxins were bound to tissues; the largest amount was in the skeletal muscle.

INDICATIONS:
Acute Infections Caused by Susceptible Strains of *Pseudomonas aeruginosa*:
Polymyxin B Sulfate is indicated for the treatment of infections of the urinary tract, meninges, and blood stream, caused by susceptible strains of Ps. aeruginosa.

It may be indicated in serious infections caused by susceptible strains of the following organisms, when less potentially toxic drugs are ineffective or contraindicated.
* H. influenzae, specifically meningeal infections.
* Escherichia coli, specifically urinary tract infections.
* Aerobacter aerogenes, specifically bacteremia.
* Klebsiella pneumoniae, specifically bacteremia.

NOTE: IN MENINGEAL INFECTIONS, POLYMYXIN B SULFATE SHOULD BE ADMINISTERED ONLY BY THE INTRATHECAL ROUTE.

To reduce the development of drug-resistant bacteria and maintain the effectiveness of polymyxin B and other antibacterial drugs, polymyxin B should be used only to treat or prevent infections that are proven or strongly suspected to be caused by susceptible bacteria. When culture and susceptibility information are available, they should be considered in selecting or modifying antibacterial therapy. In the absence of such data, local epidemiology and susceptibility patterns may contribute to the empiric selection of therapy.

CONTRAINDICATIONS:
This drug is contraindicated in persons with a prior history of hypersensitivity reactions to the polymyxins.

DOSAGE & ADMINISTRATION:
Intravenous:
Dissolve 500,000 Units Polymyxin B Sulfate in 300 to 500ml of 5% dextrose in water for continuous intravenous drip.
Adults and Children: 15,000 to 25,000 units/ kg body weight / day in individuals with normal kidney function. This amount should be reduced from 15,000 units / kg downward for individuals with kidney impairment. Infusions may be given every 12 hours; however, the total daily dose must not exceed 25,000 units / kg / day.
Infants: Infants with normal kidney function may receive up to 40,000 units / kg / day without adverse effects.
Intramuscular:
Not recommended routinely because of severe pain at injection sites, particularly in infants and children. Dissolve 500,000 Units Polymyxin B Sulfate in 2ml sterile distilled water (Sterile Water for Injection U.S.P.) or sterile physiologic saline (Sodium Chloride Injection U.S.P.) or 1% procaine hydrochloride solution.
Adults and Children: 25,000 to 30,000 units / kg / day. This should be reduced in the presence of renal impairment. The dosage may be divided and given at either 4 or 6 hour intervals.
Infant: Infants with normal kidney function may receive up to 40,000 units / kg / day without adverse effects.
Note : Doses as high as 45,000 units / kg / day have been used in limited clinical studies in treating prematures and newborn infants for sepsis caused by Ps. aeruginosa.
Intrathecal:
A treatment of choice for Ps. aeruginosa meningitis.
Dissolve 500,000 Units Polymyxin B Sulphate in 5ml of sterile physiological saline (0.9% Sodium Chloride Injection U.S.P.) for 100,000 Units per ml. Dilute further using 5ml of same solvent to obtain 50,000 Units per ml dosage units.
Adults and Children Over 2 Years of Age: Dosage is 50,000 Units once daily intrathecally for 3 to 4 days, then 50,000 Units once every other day for at least 2 weeks after cultures of the cerebrospinal fluid are negative and sugar content has returned to normal.
Children Under 2 Years of Age: 20,000 Units once daily, intrathecally for 3 to 4 days or 25,000 Units once every other day. Continue with a dose of 25,000 Units once every other day for at least 2 weeks after cultures of cerebrospinal fluid are negative and sugar content has returned to normal.

Size : L x H = 96 x 200 mm.

Pantone Process Black C

Paper : 40 gsm, ITC Print Paper
Outline & Cutting marks not to print
Artwork Code No. : IN90138E4MY
Back to Back Printing
3 Horizontal Folds

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IN THE INTEREST OF SAFETY, SOLUTIONS FOR PARENTERAL USE SHOULD BE STORED UNDER REFRIGERATION AND ANY UNUSED PORTION SHOULD BE DISCARDED AFTER 72 HOURS.

WARNINGS :

When this drug is given intramuscularly and / or intrathecally, it should be given only to hospitalized patients, so as to provide constant supervision by a physician.
Patients with nephrotoxicity due to POLY-MxB usually show albuminuria, cellular casts, and azotemia. Diminishing urine output and a rising BUN are indications for discontinuing therapy with this drug.
Neurotoxic reactions may be manifested by irritability, weakness, drowsiness, ataxia, peripheral paresthesia, numbness of the extremities and blurring of vision. These are usually associated with high serum levels found in patients with impaired renal function and / or nephrotoxicity.
The neurotoxicity of POLY-MxB can result in respiratory paralysis from neuromuscular blockade, especially when the drug is given soon after anesthesia and / or muscle relaxants.

PRECAUTIONS:
Baseline renal function should be done prior to therapy, with frequent monitoring of renal function and blood levels of the drug during parenteral therapy.
Avoid concurrent use of a curariform muscle relaxant and other neurotoxic drugs (ether, tubocurarine, succinylcholine, gallamine, decamethonium, and sodium citrate) which may precipitate respiratory depression. If signs of respiratory paralysis appear, respiration should be assisted as required, and the drug discontinued. As with other antibiotics, use of this drug may result in overgrowth of nonsusceptible organisms, including fungi. If superinfection occurs, appropriate therapy should be instituted.

DRUG INTERACTIONS:
Other antibiotics, such as aminoglycosides, can increase the potential of polymyxins for nephrotoxicity and neurotoxicity and they should be co-administered with the greatest caution with polymyxins. The concurrent use of polymyxins with a curariform muscle relaxant and other neurotoxic drugs such as ether, tubocurarine, succinylcholine, gallamine, decamethonium, and sodium citrate should be used with great caution also, since these agents may potentiate the development of neuromuscular blockade. Co-administration of sodium cephalothin and colistin may enhance the development of nephrotoxicity, so this medication should be avoided.
In addition, antimicrobial agents with known neurotoxic effect, such as aminoglycosides, should generally be in such instance, close monitoring of the patients received these antibiotics is mandatory. Experimental studies showed that application of polymyxins in combination with glutamic acid to a peripheral nerve could cause transgaglionic degenerative atrophy.

USAGE IN PREGNANCY AND LACTATION:
Pregnancy:
The safety of this drug in human pregnancy has not been established.
Lactation:
Caution should be exercised while administering Polymyxin B to lactating women as it is not studied if the drug is secreted in milk.

ADVERSE EFFECTS:
Nephrotoxic reactions:
* Albuminuria,
* Cylindruria,
* Azotemia,
* Rising serum drug levels without any increase in dosage.

Neurotoxic reactions:
* Facial flushing,
* Dizziness progressing to ataxia,
* Drowsiness,
* Peripheral paresthesias: circumoral and stocking-glove,
* Apnea due to concurrent use of curariform muscle relaxants, other neurotoxic drugs, or inadvertent overdosage,
* Signs of meningeal irritation with intrathecal administration, e.g. fever, headache, stiff neck and increased cell count and protein in cerebrospinal fluid.

Other reactions occasionally reported:
* Drug fever,
* Urticarial rash,
* Hyperpigmentation of Skin
* Pain (severe) at intramuscular injection sites,
* Thrombophlebitis at intravenous injection sites.

OVERDOSE & TREATMENT:
Overdoses with polymyxins, mainly with colistimethate sodium, have been reported several times in the old literature. Although, one case of a three year old child who received intramuscularly 450 mg (approximately 5.5 million IU) of colistimethate sodium reported no adverse effects, the majority of cases with Polymyxin overdose resulted in acuter renal failure and various manifestations of neurotoxicity, including neuromuscular blockade and apnea. It should be emphasized that cases of Polymyxin overdose with fatal consequences are scarce. There is no antidote for Polymyxin overdose. Management requires early cessation of the medication and appropriate supportive treatment. In the presence of established acute renal failure, haemodialysis and peritoneal dialysis can only manage renal complications, since they have little influence on the elimination of polymyxins, as discussed above. If apnea occurs, mechanical ventilation support is needed.

INCOMPATIBILITIES:
The concurrent or sequential use of the other neurotoxic and / or nephrotoxic drugs with polymyxin b sulfate, particularly bacitracin, streptomycin, neomycin, kanamycin, gentamicin, tobramycin, amikacin, cephaloridine, paromoycin, viomycin, and colistin should be avoided.
The neurotoxicity of polymyxin b sulfate can result in respiratory paralysis from neuromuscular blockade, especially when the drug is given soon after anesthesia and / or muscle relaxants.

STORAGE: Before reconstitution: Store below 30°C, protect from light.

After reconstitution: Product must be stored under refrigeration, between 2° and 8°C and any unused portion should be discarded after 72 hours.

PRESENTATION: 1 vial containing 500,000 Units of Polymyxin B Sulphate in a carton.

Keep medicine out of reach of children.
Jauhkan daripada capaian kanak-kanak

Date of Revision: February 2024.

BSV
Manufactured by :
BHARAT SERUMS AND VACCINES LIMITED
Plot No. K-27, K-27 Part and K-27/1, Anand Nagar, Jambivili Village,
Additional MIDC, Ambernath (East), Thane 421506, Maharashtra State, India.
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